



A cautionary tale on a cautionary tale

why the OU Model captures
evolutionary processes better
than Brownian

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Dr. Jason Pienaar

Florida International / Dept. of Biology

Models based on the Ornstein–Uhlenbeck process have become standard for the comparative study of adaptation. Cooper et al. (2016) have cast doubt on this practice by claiming statistical problems with fitting Ornstein–Uhlenbeck models to comparative data. Specifically, they claim that statistical tests of Brownian motion may have too high Type I error rates and that such error rates are exacerbated by measurement error. In this note, we argue that these results have little relevance to the estimation of adaptation with Ornstein–Uhlenbeck models for three reasons. First, we point out that Cooper et al. (2016) did not consider the detection of distinct optima (e.g. for different environments), and therefore did not evaluate the standard test for adaptation. Second, we show that consideration of parameter estimates, and not just statistical significance, will usually lead to correct inferences about evolutionary dynamics. Third, we show that bias due to measurement error can be corrected for by standard methods. We conclude that Cooper et al. (2016) have not identified any statistical problems specific to Ornstein–Uhlenbeck models, and that their cautions against their use in comparative analyses are unfounded and misleading.